ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Exploring the Investigative World of Science, Invisible Living World: Beyond Our Naked Eye, Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains why one factor is changed.
2	**(c)** A is true but R is false.	A is true; R is false because animal cells lack a cell wall.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(a)** greater	12 degrees C versus 7 degrees C.
6	**(a)** 2 mm	2 x 1000 = 2000 micrometres = 2 mm.
7	**(d)** 70	40 + 30 = 70.
8	**(b)** B gets hotter than A	Thin wire has higher resistance.
9	**(b)** 50 degrees C	Subtract zero error of 3.
10	**(c)** stops functioning properly	Nucleus controls the cell.
11	**(b)** colds are viral and often recover by themselves	Colds go away without them.
12	**(c)** the iron rod	Iron concentrates the field.
13	**(a)** Both A and R are true and R is the correct explanation of A.	Equal thickness is a controlled variable; R is the rule of a fair test.
14	**(a)** Both A and R are true and R is the correct explanation of A.	The compartments he saw were the walls of dead plant cells, which is why they looked like small rooms.
15	**(d)** A is false but R is true.	A is false because health is complete well-being; R is true.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	T = 34 / 20 = 1.7 s (1); Timing many swings reduces the error of reaction time (1).	1+1
17	Actual = 5 / 500 = 0.01 mm (1); = 10 micrometres (1).	1+1
18	Unvaccinated = 2000 = 25% (1); Fewer susceptible persons means less spread (1).	1+1
19	Both feet are at the same potential (1); No current passes through the bird (1).	1+1
20	Accuracy is closeness to the true value; precision is closeness of repeated readings (1); Readings may be precise but wrong due to zero error (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Only one soil type was compared (1); The conclusion over-generalises (1); Test several soils with controls (1).	1+1+1
22	Onion cells are rectangular with a cell wall; cheek cells are rounded or irregular without a wall (1); Both show a nucleus and cytoplasm; the onion cell has a large vacuole (1); Stains such as safranin or methylene blue make the parts visible (1).	1+1+1
23	Contaminated drinking water (1); Unhygienic food (1); Preventive: boiled water, hygiene, treat water tank (1).	1+1+1
24	Increase the number of turns (1); Increase the number of cells (1); Use a thicker soft iron core (1).	1+1+1
	OR: Straight line (1); Passes through the origin (1); Ratio stays constant (1).	1+1+1
25	Digestion: food vacuole (1); Excretion: contractile vacuole (1); Movement: pseudopodia (1).	1+1+1
	OR: Goitre: iodised salt (1); Rickets: milk, sunlight (1); Anaemia: leafy vegetables (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) Amount of water in the dough (1). (ii) Time taken to puff (or degree of puffing) (1). (iii) Quantity of flour / thickness of puri / oil temperature (any one) (1). (iv) Hypothesis: puris made with more water puff in less time (or puff better) (1). Repeating trials checks for chance errors and makes the result reliable (1).	1+1+1+2
27	Contaminated food or water at the feast (1); Salmonella bacteria from a carrier or dirty water (1); Hands and flies spread germs (1); Ensure clean water, cooked food, hygiene (1); Treatment with antibiotics on advice; vaccination (1).	1+1+1+1+1
	OR: Independent: number of turns; dependent: pins lifted (1); Constant: cell, nail, wire (1); Procedure: 10, 20, 30, 40 turns and count pins (1); Expected: pins rise with turns (1); Conclusion (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Exploring the Investigative World of Science, Invisible Living World: Beyond Our Naked Eye, Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Series connection allows it to sense all current.
2	**(a)** B and C	Only light differs between B and C.
3	**(b)** 0.05 mm	5 / 100 = 0.05 mm.
4	**(b)** 75%	15000/20000 x 100 = 75%.
5	**(b)** glows dimmer	Reversed cell opposes the others.
6	**(a)** plants without fertiliser grown in same conditions	Needed for comparison.
7	**(a)** yeast is more active at warm temperature	Warmth boosts fermentation.
8	**(b)** boiling milk	Not related to vector.
9	**(a)** the current is larger	More current.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Only testable statements can be supported or rejected.
11	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
12	**(a)** Both A and R are true and R is the correct explanation of A.	Breeding sites are removed.
13	**(a)** Both A and R are true and R is the correct explanation of A.	Oersted's observation is explained by the magnetic field of the current.
14	**(a)** Both A and R are true and R is the correct explanation of A.	R is the definition that explains why A holds.
15	**(a)** Both A and R are true and R is the correct explanation of A.	The cell wall gives the plant cell its fixed shape.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Amoeba has only a flexible membrane (1); Plant cell has a rigid wall (1).	1+1
17	They cannot be cured completely (1); Treatment and lifestyle changes keep them in control (1).	1+1
18	Total resistance increases (1); Less current for each (1).	1+1
19	Numbers and units are clear at a glance (1); Comparison and plotting are easier (1).	1+1
20	All organisms are built of cells (1); Cells combine into tissues, organs and bodies (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Wire resistance is high (1); Less current reaches the bulb (1); Use a short thick wire (1).	1+1+1
22	Equal size sheets, same water volume (1); Measure water absorbed (1); Repeat and average (1).	1+1+1
23	Antibiotics act on bacteria (1); Fungi have different cell walls (1); Fungicides are needed (1).	1+1+1
24	Clean water and mess hygiene (1); Regular medical check-ups (1); Mosquito control and cleanliness (1).	1+1+1
	OR: The switch breaks the circuit (1); No current flows (1); But switching off the plug is also advised (1).	1+1+1
25	Outlier: 6.5 s (1); Cause: late stop or a different path (1); Mean without it = 16.4 / 4 = 4.1 s (1).	1+1+1
	OR: Virus is smaller than bacterium (1); Virus is non-cellular; bacterium is a cell (1); Bacterial diseases need antibiotics; viral diseases need vaccines/antivirals (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) A thin peel lets light pass through so one layer of cells can be seen (1). (ii) Nucleus (1). (iii) Cell wall (1). (iv) Cheek cells have no cell wall and are irregular/rounded in shape, and have no large central vacuole (any two) (2).	1+1+1+2
27	Pathogen causes disease; host harbours it (1); Vector carries it (1); Symptoms felt, immunity resists (1); Vaccine prevents; antibiotic treats bacteria (1); Epidemic is a sudden spread (1).	1+1+1+1+1
	OR: Cell about 1.5 V (1); Battery: several cells, higher voltage (1); Mains about 220 V (1); Low voltage cannot drive a harmful current through the body (1); Mains can cause fatal shock (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Exploring the Investigative World of Science, Invisible Living World: Beyond Our Naked Eye, Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** The pathogen can spread through unvaccinated people	Community protection weakens.
2	**(d)** strength rises with turns	Doubling turns gives about double pins.
3	**(a)** time for wax to drop	Result measured.
4	**(b)** tissues	Tissue means many cells.
5	**(c)** isolating the patients and hygiene	Break transmission.
6	**(a)** retains magnetism	Steel is used for permanent magnets.
7	**(a)** Both A and R are true and R is the correct explanation of A.	Honest recording of all data is needed.
8	**(a)** Both A and R are true and R is the correct explanation of A.	The space allows more haemoglobin.
9	**(d)** A is false but R is true.	A is false: diabetes is non-communicable. R is true: it is not caused by a pathogen.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Oersted's finding.
11	**(c)** 1.6 s	48 / 30 = 1.6 s.
12	**(b)** virus, bacterium, red blood cell	Smaller to larger.
13	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; R describes how the mosquito becomes infected, and A states the route to a new person, so R does not directly explain A.
14	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the loss of magnetism on switching off does not explain why the magnet is stronger.
15	**(d)** A is false but R is true.	A is false because repeating readings improves reliability; R is true.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Prevent contamination of water (1); Reduce fly and mosquito breeding (1).	1+1
17	Current decreases (1); The electromagnet becomes weaker (1).	1+1
18	Sum of five = 62.0 (1); Fifth = 62.0 - 49.4 = 12.6 s (1).	1+1
19	Cork consists of dead cells (1); Only the walls remain after the contents are lost (1).	1+1
20	Flu is viral so antibiotics do not work (1); Rest and fluids help the immune system (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Vaccine prevents infection; does not cure (1); Medicine treats symptoms (1); Measles has no antibiotic cure as it is viral (1).	1+1+1
22	The heater (1); It carries a larger current (1); A thick wire has low resistance and does not overheat (1).	1+1+1
23	Wanting a certain result can influence readings (1); Use blind or double checks (1); Record all readings honestly (1).	1+1+1
24	No division of labour (1); Complex functions cannot be done (1); Organisms would stay simple (1).	1+1+1
	OR: Individual: repellent, long sleeves (1); Family: nets, no stagnant water (1); Community: fogging, drain cleaning (1).	1+1+1
25	Both give deflection since current flows (1); Iron foil may change the field slightly (1); Paper has no effect (1).	1+1+1
	OR: Experiment changes a variable to see effect (1); Survey collects information without changing anything (1); e.g. fertiliser test versus asking students' favourite fruit (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) Malaria (1). (ii) Female Anopheles mosquito (1). (iii) A protozoan (Plasmodium) parasite (1). (iv) Do not let water stagnate (empty coolers, tyres, pots weekly); use mosquito nets/repellents; spray insecticide or oil on ponds (any two) (2).	1+1+1+2
27	First dose: slow, small rise (1); Second dose: faster, higher rise (1); Memory cells remain (1); Levels fall later (1); Booster restores (1).	1+1+1+1+1
	OR: Total = 4.5 + 0.5 + 0.3 (1); = 5.3 A (1); Exceeds 5 A, so unsuitable (1); Fuse would melt in use (1); Choose a 10 A fuse for safety (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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